Windows logon types & table with meaning and one normal vs one suspicious example.

Logon Types are important fields within Windows authentication events, particularly Event ID 4624 and Event ID 4625.

They describe how a user or account accessed a Windows system.
Understanding these types helps SOC analysts identify suspicious authentication behaviour.

1. **Logon Type comparison table**

Logon Type	Meaning	Normal example	Suspicious example
2	Interactive	Employee logs in directly at their office computer.	An employee's account logs in locally on a server at 3 AM without an approved reason.
3	Network	User accesses a shared folder on a file server.	An unknown workstation attempts to authenticate to multiple servers using an administrator account.
4	Batch	Scheduled backup task runs under a service account.	An unexpected scheduled task runs under a privileged account.
5	Service	Windows service starts using its configured service account.	An unknown service starts using an unexpected privileged account.
7	Unlock	Employee unlocks their workstation after lunch.	An account repeatedly unlocks a workstation unexpectedly or at unusual times.
10	RemoteInteractive	Employee connects to their workstation using Remote Desktop.	An administrator account connects through RDP from an unknown external IP.
11	CachedInteractive	Employee logs in using cached domain credentials while disconnected from the domain controller.	A user account is used on a stolen laptop while disconnected from the corporate network.

2. **Detailed explanation of each logon type**
   i. **Logon Type 2 - Interactive Logon**
   This occurs when a user logs on directly to a computer.

   Example:
   An employee enters their username and password on their office desktop.

   Normal:

Event ID: 4624
Logon Type: 2
Account: john
Workstation: OFFICE-PC-01

Suspicious:
A standard employee account logs on locally to a domain controller without a legitimate administrative reason.

What to investigate:

- Account identity.
- Device name.
- Time of access.
- Whether the user normally accesses that system.

ii. Logon Type 3 - Network Logon

This occurs when a user accesses a computer over a network.
Example:
A user accesses a shared folder:
\\FILESERVER\Shared

Normal:
An employee accesses a shared folder from their regular workstation.

Suspicious:
An unknown workstation attempts to access ADMIN$ shares across many systems.

Common associated activities:

- SMB access.
- Network authentication.
- Remote resource access.

Important: Network logons do not automatically mean Remote Desktop access. RDP generally uses Logon Type 10.

iii. Logon Type 4 - Batch Logon

This occurs when a user or account logs on to execute a batch job.

Example:
A scheduled backup task runs every night at 2 AM.

Normal:
Account: backup_service

Logon Type: 4
Time: 02:00

Suspicious:
An unexpected script runs under a privileged user account outside the approved schedule.

What to investigate:

- Scheduled tasks.
- Task configuration.
- Script paths.
- Service and task accounts.
- Process creation events.

iv. Logon Type 5 - Service Logon

This occurs when the Service Control Manager starts a Windows service using its configured account.

Normal:
A Windows service starts using Local System.

Suspicious:
An unknown service runs under a domain administrator account.

What to investigate:

- Service name.
- Executable path.
- Service account.
- Service installation events.
- Process creation events.

v. Logon Type 7 - Unlock

This occurs when a previously locked workstation is unlocked.

Normal:
An employee returns from lunch and unlocks their computer.

Suspicious:
A workstation is repeatedly unlocked using an unexpected account.

What to investigate:

- User identity.
- Workstation.
- Timing.

- Corresponding lock events.
- Whether the account owner was present.

vi. Logon Type 10 - RemoteInteractive

Important for RDP investigations
This generally represents a Remote Desktop or Terminal Services interactive logon.

Example:
An employee remotely connects to an office computer using RDP.

Normal:
An authorized IT administrator connects from a corporate VPN address.

Suspicious:
An administrator account connects from an unfamiliar public IP address outside business hours.

What to investigate:

- Source IP.
- Account name.
- RDP authorization.
- VPN logs.
- MFA status.
- Event ID 4624.
- Event ID 4625.
- Event ID 4634.

vii. Logon Type 11 - CachedInteractive

This occurs when Windows authenticates a domain user using cached credentials because a domain controller is unavailable.

Normal:
An employee logs into a corporate laptop while travelling without internet access.

Suspicious:
A stolen corporate laptop is used offline with a cached domain account.

What to investigate:

- Device location.
- Last known network activity.
- Device security status.
- User account status.
- Subsequent online authentication.

3. Quick memory trick

Type	Remember it as
2	Physical/local login
3	Network access
4	Scheduled batch job
5	Service startup
7	Unlock workstation
10	Remote Desktop
11	Cached domain login